

BDAT04 — PRELIMINARY ASSESSMENT · TEST II PRACTICAL DEMONSTRATION

Botika Maligaya dataset · 45 points · 120 minutes · draw ONE task from each band

One task per sheet. Write your EVIDENCE and DECISION on this sheet and hand it in.

Name: Lagdamen, Lawrence M. Drawn at: 10:50 AM Finished: 11:09 AM

BAND A · TASK A1 · 8 POINTS

A single operation with one correct outcome

What did you just load?

Load the main sales file. Report the number of rows, the number of columns, and the data type Power Query has assigned to QuantityDispensed. In one sentence, say why that type is a problem for a fact table.

EVIDENCE: *(the counted result, visible on screen)*

Number of Rows: **6024**

Number of Columns: **9**

QuantityDispensed data type: Whole Number

DECISION: *(written, in full sentences)*

The data type of QuantityDispensed is correct as it reports the quantity of the medicine that a specific branch has dispensed. However, the Date column has a false data type and should be adjusted accordingly.

Applied steps renamed? YES Query name: Date change to "Date" data type

Name: Lagdamen, Lawrence M. Drawn at: 11:10 AM Finished: 11:26 AM

BAND A · TASK A2 · 8 POINTS

A single operation with one correct outcome

Prove a column is clean

Load the main sales file. Count the distinct values in PaymentMethod and list each one with its row count. State in one sentence whether this column needs cleaning.

EVIDENCE: *(the counted result, visible on screen)*

Card: 790

Cash: 3184

Gcash: 1462

HMO: 588

DECISION: *(written, in full sentences)*

This column does not need to be cleaned. All data are properly accounted for.

Applied steps renamed? NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 11:26 AM Finished: 11:30 AM

BAND A · TASK A3 · 8 POINTS

A single operation with one correct outcome

A blank that may not be a defect

Load the main sales file. Count how many rows have a blank SeniorCitizenID and express it as a percentage of all rows. State in one sentence whether this is a data quality defect.

EVIDENCE: *(the counted result, visible on screen)*

SeniorCitizenID null rows: 262 rows or 4.35% of the data

DECISION: *(written, in full sentences)*

This is not a data quality defect, it simply means that there are 262 customers who are not Senior Citizens.

Applied steps renamed? NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 11:30 AM Finished: 11:41 AM

BAND B · TASK B1 · 10 POINTS

Two steps where the order matters

The column that will not convert

QuantityDispensed will not convert to a whole number. Repair it so the column sums. Report how many values contained a comma, and the total quantity dispensed across the file.

EVIDENCE: *(the counted result, visible on screen)*

QuantityDispensed with a comma: 17

QuantityDispensed negative counts which suggest spoilage: 4

DECISION: *(written, in full sentences)*

There are a total of 17 rows of data with commas.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 11:41 AM Finished: 11:48 AM

BAND B · TASK B2 · 10 POINTS

Two steps where the order matters

Append, then dedupe

Append the July 2026 sales file to the main sales query, then remove duplicate rows. Report the row count AFTER EACH STEP, and how many rows the deduplication removed.

EVIDENCE: *(the counted result, visible on screen)*

AFTER APPEND: 6544

AFTER REMOVING DUPLICATE ROWS: 34

DECISION: *(written, in full sentences)*

A total of 6510 rows were removed

Applied steps renamed? YES / NO Query name: _____

BAND B · TASK B3 · 10 POINTS

Two steps where the order matters

More branches than exist

BranchID shows more branches than the branch list contains. Repair the formatting variants — delete nothing — then report how many distinct BranchID values remain and how many rows you repaired.

EVIDENCE: *(the counted result, visible on screen)*

PH-02: 8

PH-05: 6

(ph-01): 11

(ph-04): 6

PH03: 7

DECISION: *(written, in full sentences)*

I repaired a total of 38 rows, resulting in 6024 rows in total, whereas there were 6544 initially.

Applied steps renamed? YES Query name: Standardized BranchID

Name: Lagdamen, Lawrence M. Drawn at: 12:00 NN Finished: 12:09 NN

BAND C · TASK C1 · 12 POINTS

A diagnostic — find it, count it, say what it costs

Products that do not exist

Using a left anti join, find every sales row whose ProductID does not appear in the product list. Count them, name the value or values, and state what happens to those rows in a report broken down by product category.

EVIDENCE: *(the counted result, visible on screen)*

Number of ProductID who does not appear in the product list: 11 Rows.

Product ID: DRG999

DECISION: *(written, in full sentences)*

It should be noted that there are orphan rows in the ProductID, and they should be investigated further.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 12:10NN Finished: 12:13NN

BAND C · TASK C2 · 12 POINTS

A diagnostic — find it, count it, say what it costs

Branches that do not exist

Using a left anti join, find every sales row whose BranchID does not appear in the branch list. Count them, then SEPARATE the ones that are a formatting problem from the ones that are genuinely unknown. Report both counts.

EVIDENCE: *(the counted result, visible on screen)*

Total number of Rows: 29

Formatting problem: 7

Unknown: 22

DECISION: *(written, in full sentences)*

Among the BranchIDs there are a total of 29 orphan rows, 7 of which are formattin errors, and 22 of which are genuinely unknown

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 12:13NN Finished: _____

BAND C · TASK C3 · 12 POINTS

A diagnostic — find it, count it, say what it costs

The number that does not reconcile

Compare TotalAmount against QuantityDispensed × UnitPrice. Count the rows that disagree, and SPLIT the disagreement by direction. State which direction has an innocent explanation and which does not.

EVIDENCE: *(the counted result, visible on screen)*

Rows with counts of 1: 5980

DECISION: *(written, in full sentences)*

When the check was done, multiple decimal values were present. While there are true values of 5980 rows, the others suggest the spoilage or the comma formatting error.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 12:20NN Finished: _____

BAND D · TASK D1 · 15 POINTS

A judgement — decide, and defend it in writing

The orders that look wrong

After repairing QuantityDispensed, run an outlier test on it. Report the upper fence and how many rows exceed it. Decide what to do with those rows — fix, exclude, escalate or accept — and defend the decision in writing.

EVIDENCE: *(the counted result, visible on screen)*

21 Values

33 rows in total

DECISION: *(written, in full sentences)*

Escalate, these data do not necessarily mean that they are offending values. They could mean that a specific Branch has sold more products.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: 12:25NN Finished: _____

BAND D · TASK D2 · 15 POINTS

A judgement — decide, and defend it in writing

The negative numbers

QuantityDispensed contains negative values. Count them, state what they represent, and decide: fix, exclude, escalate or accept. Defend the decision in writing.

EVIDENCE: *(the counted result, visible on screen)*

Rows in total: 15

DECISION: *(written, in full sentences)*

Escalate, the values represent spoilage.

Applied steps renamed? YES / NO Query name: _____

Name: Lagdamen, Lawrence M. Drawn at: Finished: 12:31NN

BAND D · TASK D3 · 15 POINTS

A judgement — decide, and defend it in writing

The month with nowhere to land

You have appended the July 2026 sales file. Test the sales dates against the calendar table. Report what you find, and decide what must be done before this model is refreshed again. Defend the decision in writing.

EVIDENCE: *(the counted result, visible on screen)*

Null data: 9

DECISION: *(written, in full sentences)*

Escalate, this data could mean that they are duplicates.

Applied steps renamed? YES / NO Query name: _____